双语全文翻译｜From GIS to neogeography ontological implications and theories of truth

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REVIEW ARTICLE
From GIS to neogeography: ontological implications and theories of truth
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Neogeography has emerged as a descriptive and analytical tool for large numbers of people outside of academia, a process catalyzed by digital mapping technologies and the social networking practices of W eb 2.0. This article examines the ontological and epistemological implications of this transition. It argues that neogeography has democratized GIS practices, facilitated the adoption of relational views of space and place, and broadened the ontological scope of GIS. Neogeography also poses epistemological challenges to the dominant theory of truth, in particular advancing a shift away from the correspondence model of truth toward consensus and performative interpretations. While recognizing the multiple unintended consequences of neogeographical practices, neogeography is held to be as a useful means for charting geographic space in light of intense postmodern time–space compression.
Keywords: neogeography; ontology; theories of truth; relational space; crowdsourcing; volunteered geographic information
W eb 2.0

Introduction

The dazzling, sometime even dizzying, development of new technologies for handling geospatial information has caught everybody by surprise over the past decade, particularly in the last 5 years. Neologisms burst into our lexicon at an overwhelming rate: W eb 2.0, user-generated content, participatory GIS, 'do-it-yourself ' cartography, MapQuest, Bing maps, location-based services, cloud computing, wikification, Google Earth, blogging, Flickr, digiplaces, Y ouTube, Geocaching, Second Life, augmented reality, T witter, and Foursquare. All of these phenomena point to a profound shift in the nature and role of geographic information, a transition characterized by a 'bottom up' reconfiguration in how data are collected, transmitted, analyzed, visualized, and utilized that differs considerably from traditional 'top-down' models in which experts and government agencies dictate the criteria of data collection, analysis, applications, and standards of truth.

These diverse geographical practices have been loosely called neogeography (Turner 2006), a term coined by Di-Ann Eisnor of Platial (www.platial.com) (Walsh 2008). As Haden (2008) noted, the term has been used multiple times at major technological or epistemological junctures in geography throughout the twentieth century. W e adopt both a narrow and a broad definition for neogeography (V ander Wal 2005). A narrow view refers to the process whereby varied groups of people use an eclectic set of

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online geospatial tools to describe and document aspects of their lives, society, or environment in terms that are meaningful to them (Hudson-Smith et al . 2009). Defined this way, neogeographical knowledge emanates from personal interactions with places, and what is held to be true is often highly contextual and specific to the community of users who generate such data. In a sense, then, neogeographical knowledge is inseparable from the rhythms and contours of everyday life. Rana and Joliveau (2009, p. 75) argue that to fully appreciate the significance of this phenomenon, neogeography must be regarded 'as an extension of mainstream geography for everyone made by everyone.' A broader interpretation of neogeography includes the cultural mapping of all realms of everyday life, catalyzed by the digital mapping technologies and social networking practices of W eb 2.0 (Barnes and Sheppard 2010). As

Goodchild (2009a) points out, neogeography often implies that academic geography is redundant or unnecessary. For this reason, he instead advocates terms such as volunteered geographic information (VGI).

In many ways, as demonstrated by the papers presented during the 2010 neogeographers' annual meeting, the Where 2.0 Symposium (http: //en.oreilly.com/where2010), neogeography, code spaces, and the automatic production of space have become deeply woven into the fabric of daily life for countless millions of people to such an extent that simple dichotomies like 'online' and

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新地理学已经成为学术界之外大量人群进行描述和分析的工具；数字制图技术以及 Web 2.0 的社交网络实践催化了这一过程。本文考察这种转变的本体论和认识论含义。文章认为，新地理学使 GIS 实践民主化，促进了关系性的空间与地方观念，并拓宽了 GIS的本体论范围。新地理学也向占主导地位的真理理论提出认识论挑战，尤其推动人们从真理的符合论转向共识论和施为论解释。尽管承认新地理实践会产生多种意料之外的后果，本文仍认为，在后现代时空压缩日益强烈的背景下，新地理学是描绘地理空间的有用手段。

关键词：新地理学；本体论；真理理论；关系空间；众包；志愿地理信息；Web 2.0。引言。过去十年，特别是最近五年，用于处理地理空间信息的新技术发展令人眼花缭乱、甚至目眩神迷，出乎所有人的意料。新词以压倒性的速度涌入我们的词汇：Web 2.0、用户生成内容、参与式 GIS、“ 自己动手”制图、MapQuest、必应地图、基于位置的服务、云计算、维基化、Google Earth、博客、Flickr、数字地方、YouTube、地理藏宝、第二人生、增强现实、Twitter 和 Foursquare。所有这些现象都指向地理信息性质与作用的深刻转变：数据的收集、传输、分析、可视化和使用方式正在“ 自下而上”地重构，与传统的“ 自上而下”模式大不相同；在传统模式中，专家和政府机构规定数据收集、分析、应用以及真理标准。这些多样的地理实践被笼统地称为“ 新地理学”（Turner 2006），该术语由 Platial 的 Di-Ann Eisnor 首创。我们采取狭义和广义两种定义。狭义上，新地理学是不同群体利用各种在线地理空间工具，以对自己有意义的方式描述和记录自身生活、社会或环境的过程。这样定义的新地理知识来自个人与地方的互动；何者被视为真实，往往高度依赖情境，并具体对应于生成数据的用户社群。因此，新地理知识与日常生活的节奏和轮廓不可分离。Rana 和 Joliveau 认为，要充分理解这一现象，必须把新地理学看作“ 人人为人人制作的主流地理学延伸”。广义解释则包括对日常生活所有领域的文化制图，其动力来自 Web 2.0 的数字制图技术和社交网络实践。Goodchild 指出，新地理学常常暗示学术地理学是多余或不必要的，因此他主张使用“ 志愿地理信息”（VGI）等术语。正如 2010 年新地理学家年度 Where 2.0 研讨会上的论文所展示的那样，新地理学、代码空间和空间的自动生产已经深深织入数以亿计的人们的日常生活，以至于“ 在线”与“ 离线”这类简单二分无法公正描述现实世界与虚拟世界相互渗透的方式。 Web 2.0 还催生了用户生成内容、参与式 GIS、“ 自己动手”制图、MapQuest、Bing Maps、基于位置的服务、云计算、维基化、Google Earth、Flickr、数字地方、地理藏宝、第二人生、增强现实、Twitter 和 Foursquare 等一系列词汇。新地理学一词在 20 世纪地理学经历重大技术或认识论转折时曾多次出现；本文采用 Vander Wal 所说的狭义与广义定义。狭义定义强调在线工具由不同群体为记录自身生活而使用，知识来自个人与地方的接触；广义定义则涵盖 Web 2.0 促成的日常生活各领域文化制图。2010 年 Where 2.0 研讨会的成果表明，代码空间和空间自动生产已融入数以百万计的人们的日常生活，因此“ 在线 / 离线”的二分无法说明现实与虚拟世界相互渗透的程度。

‘off-line’ fail to do justice to the ways in which the real and the virtual worlds are interpenetrated (Thrift and French 2002, Dodge and Kitchin 2005, Gordon 2008, Messinger et al . 2009). Although this phenomenon has been the subject of much recent scrutiny (Goodchild 2007a, b, 2009b, Elwood 2008, Coleman et al . 2009, Coote and Rackham 2009, Batty et al. 2010, Haklay 2010), the ontological and epistemological implications have remained largely unexplicated. Although one philosophical assessment has been offered about W eb 2.0 and the semantic web in general (Floridi 2009), as yet there exists little systematic theoretical scrutiny of neogeography, which is supported by W eb 2.0 technologies and the development of the semantic web. The aim of this article is to clarify the complex, multifaceted implications of neogeography for the production of geographical knowledge in general, and theories of knowledge and truth in particular, at both the ontological and epistemological levels. In an age in which theoretical self-consciousness and sophistication have reached unparalleled heights among critical social theorists, GIScientists can ill-afford to view the creation of ‘truth’ unproblematically when it has increasingly come to have a radically different meaning from the conventional academic one.

The article consists of four parts. It begins with a recapitulation of critical human geography’ s objections to conventional GIS, that is, its obsession with mechanical rationality, its denial of an observer and reliance on the correspondence theory of truth, its assumption that representation is only a technical, not social process, and the view of absolute space embedded within such approaches. Second, it turns to the rise of neogeography and critical GIS and its implications for data /information gathering and ultimately for knowledge production. W e highlight how in practice neogeographers use geospatial technologies in multiple ways that are consistent epistemologically with the views of critical GIS scholars. Third, it discusses the pluralist ontologies that neogeography espouses, that is, the multitudinous realities perceived and inhabited by different social groups, with particular emphasis on representing relative and relational space. Fourth, we examine new approaches to truth and how these different approaches can be applied to neogeographies, namely, ways of understanding space that are free from the unrealistic constraints of the correspondence theory of truth, drawing on the works of three philosophers, Paul Feyerabend, Jürgen Habermas, and Richard Rorty. W e discuss how neogeographical practices are consistent with alternative theories of truth such as the consensus or performative theories of truth. The last section offers a conclusion that situates neogeographic practices in the context of fractalized, postmodern life.

Conventional versus critical GIS

To appreciate neogeography’ s implications as a new mode for geographical knowledge production, a quick review is offered to discuss the critiques of conventional GIS and how neogeography reflects the broader rise of critical GIS. Following a long tradition of critical cartography (Harley 1989, W ood 1992, Crampton and Krygier 2005), it has become common for geographers to view maps as much more than simply objects, but as configurations of power and knowledge that are indelibly linked to the social relations that produce them. Such an epistemolog-

ical shift challenges conventional views of representation as a purely technical act and recasts it as one intimately tied to social interests, that is, with social origins and social consequences. In this light, the traditional boundary between epistemology (the theory of knowledge, i.e., how do we know?) and ontology (the theory of reality, i.e., what exists?) dissolves. Maps as a form of spatial knowledge do much more than simply reflect the world, but in fact discursively constitute it, or as W ood (1992) notes, they help to bring into existence the world that they portray. Seen from this perspective, maps – and geospatial technologies in general – lose much of their conventional aura of objectivity and scientific detachment and are recast as social products with social origins and social consequences.

Conventional GIS is essentially spatially deterministic in nature in that it requires representations of landscapes and processes to be tied to the geometrical primitives of points, lines, polygons, and pixels. The mathematical topology that underpins GIS brings its own data representations in the form of raster, vector, and object forms. The deployment of these geometric forms enables conventional GIS to classify readily observable empirical phenomena such as natural resources, infrastructure, demographic patterns, and environmental events rather than use the less well-defined descriptive terms and categories deployed in critical geography and social theory such as class, ethnicity, gender, and other markers of social identity. Because technologies shape how we view the world, conventional GIS inevitably rested upon, and amplified, an essentially positivist philosophical perspective at the expense of others. Epistemologically, conventional cartography and GIS tend to privilege the quantitative and the observable over the qualitative and the nonobservable. Reality is therefore assumed to consist of only what is seen or measurable, and it is things themselves, not the relations among them, which hold ontological priority (a notion that pervades both positivism and empiricism). Such a perspective upholds a deductive, nomothetic rationality that marginalizes other logics such as social constructivism or hermeneutics. Conventional GIS unsurprisingly tends to value precision, simplification, certainty, and deterministic causality over complexity, ambiguity, multiplicity, and contingency, the very things that most heavily concern critical social theorists.

Epistemologically, conventional GIS relies exclusively on the correspondence theory of truth, particularly the version formalized by the Vienna Circle in the early twentieth

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虽然这一现象最近受到大量审视，但其本体论和认识论含义仍未得到充分阐释。本文旨在澄清新地理学对地理知识生产，以及对知识和真理理论的复杂、多层面影响，既涉及本体论层面，也涉及认识论层面。在理论自觉和复杂程度达到前所未有高度的时代，GIS科学家不能再把“ 真理”的创造视为理所当然，因为它的含义已越来越不同于传统学术意义。文章分为四个部分：首先回顾批判性人文地理学对传统 GIS的反对——对机械理性的迷恋、否认观察者并依赖符合论、假定表征只是技术过程而非社会过程，以及其中所包含的绝对空间观。第二部分讨论新地理学与批判性 GIS的兴起及其对数据 / 信息收集乃至知识生产的影响。第三部分讨论新地理学所支持的多元本体论，即不同社会群体感知并居住于众多现实之中，特别关注相对空间和关系空间的表现。第四部分考察新的真理观，以及如何把这些方法用于新地理学：借鉴保罗▲费耶阿本德、尤尔根▲哈贝马斯和理查德▲罗蒂的思想，寻找摆脱符合论不现实限制的理解空间方式。传统 GIS与批判性 GIS。要理解新地理学作为地理知识生产新模式的含义，有必要简要回顾传统 GIS 所受到的批评，以及新地理学如何反映批判性 GIS的更广泛兴起。沿袭批判制图学的传统，地理学家已普遍把地图看作远不止物件，而是权力与知识的构型，并与生产它们的社会关系不可磨灭地联系在一起。这种认识论转向挑战了把表征视为

纯技术行为的传统观念，把它重新界定为与社会利益密切相关、具有社会起源和社会后果的行为。在这一视角下，认识论（知识理论：我们如何知道？）与本体论（现实理论：存在什么？）之间的传统边界消解了。地图作为空间知识，不只是反映世界；它实际上通过话语构成世界，正如伍德所说，地图帮助把它所描绘的世界带入存在。从这个角度看，地图以及一般的地理空间技术失去了传统的客观性和科学超然光环，转而被理解为具有社会起源和社会后果的社会产品。

文章随后明确说明四部分结构：第二部分转向新地理学和批判性 GIS，分析数据 / 信息收集对知识生产的影响；第三部分讨论多元本体论及不同社会群体所感知和居住的多重现实，尤其是相对空间与关系空间；第四部分借助 Feyerabend、Habermas 和 Rorty 的著作，讨论共识论与施为论等替代理论。批判制图学的传统提醒我们，地图是权力与知识的配置，而不是中性的物体；因此表征的认识论和存在论必须同时考察。

century, in which the truth value of theoretical statements is judged by their isomorphism with observed facts, that is, the correspondence between hypotheses and data. Truth, in this view, unproblematically exists ‘out there,’ independently of whether or not anyone believes in it or not, which is to say that truth is held to be not a social construction but an objective, given fact. In the correspondence model, truth is ‘real’ whether or not anyone believes in it. Such a perspective subscribes to a strict dichotomy between theoretical and empirical languages (i.e., the data are assumed not to be theory-laden: facts are given, not made), renders the role of the observer invisible or irrelevant, and has long been dominant not only in Anglophone analytic philosophy but in much of the social sciences as well. This notion confers the power to know and define the world – both social and natural – on an invisible, rational, implicitly male, decontextualized, all-knowing observer – the Cartesian cogito. However, in the face of intense criticisms, many social scientists, and practitioners of critical GIS, have abandoned the strict positivist view of the world, although it continues to dominate the physical sciences.

In addition to its views of knowledge and the knower, conventional GIS also relies upon a particular notion of space. Specifically, like the isotropic plains of location theory and regional science, traditional practitioners of GIS invariably invoke an absolute Cartesian view of spatiality as given, not produced, a homogenous surface devoid of social origins. Absolute space – infinite, absolute, and homogenous – has a long lineage in W estern history that can be traced back to classical Greece, including geometry. Euclidean geometry, grounded in the assumption of one, uniform, continuous space, dominated the mathematics of spatial representation for the next two millennia. In Renaissance Europe, Cartesian views of space were instrumental in the development of cartography, land surveying, civil engineering, and architecture (Cosgrove 2008). The Cartesian /Euclidean notion of space was replicated throughout the Renaissance and the Enlightenment and formed the basis for Newtonian physics. Such a perspective was instrumental to the emerging capitalist logic of globalized commodity production and consumption. For European navigators, for example, smoothing space by reducing it to distance rendered the oceans navigable, and arranged the multitude of the planet’ s places into a single, unified, coherent, and panopticonic understanding of the world. In the United States, the township and range system played a similar role. In the process, space was robbed of substantive social context to become an ordered and uniform system of abstract linear coordinates (e.g., the grid of latitude and longitude).

The social constructivist critisms of conventional cartography found their way into GIS in the 1990s, with dramatic results (Pickles 1995, Sheppard 1995, 2005a, b, Schuurman 2000, 2006). Because trenchant critiques are found elsewhere, there is no need to recapitulate this literature here. Suffice it to say that critics held that conventional GIS lacked a sufficient engagement with issues of social context, relations, and power, amounting to little more than a Trojan horse for smuggling empiricism or positivism back into geographic practice. Others were concerned about the military and corporate origins and

applications of much of GIS, potential invasions of privacy, and the panopticonic implications of centralized geospatial databases (Armstrong and Ruggles 2005, Dobson and Fisher 2007). The community of conventional GIS practitioners initially amplified the schism with social theorists through its impoverished epistemological self-understanding, which was typically framed in a purely technocratic language of engineering, efficiency, control, and manipulation, as if GIS were devoid of social roots and consequences.

However, over time, in response to these criticisms, the GIS community has changed steadily. Indeed, there is no reason why GIS cannot 'be employed within non-positivist epistemologies' (Pavoloskaya 2002, p. 287). The evolution of GIS since the 1990s reveals a history that currently includes a fantastic zoo of new species, including, to take but a few examples, critical GIS (Sheppard 2005b), participatory GIS (Elwood 2006), volunteered geography (Goodchild 2007b), affective /emotional GIS (Aitken and Craine 2009), qualitative GIS (Cope and Elwood 2009), feminist GIS (Kwan 2002a, McLafferty 2002, 2005, Story 2002), queer GIS (Brown and Knopp 2008), digiplaces (Zook and Graham 2007a), ethnographic and indigenous knowledge (Robbins 2003), and humanistic GIScience (Sui 2004, Bodenhamer et al . 2010). These developments reflect a highly productive engagement between the GIS community and social theorists and humanities scholars during the last decade. Interactions and exchanges between GIScientists and social theorists stimulated the emergence of so-called critical GIS that attempts at 'pixelizing the social and socializing the pixel' (Geoghegan et al . 1998). Although there are certainly significant differences among these diverse approaches, there are also numerous similarities that wind their way through this body of work. Thus, contemporary forms of GIS, unlike those dominant in the 1990s, tend to be much more context-sensitive and issue-driven than their predecessors. In addition to paying more attention to issues of power, commodification, and surveillance, GIS practices nowadays are also more likely to tolerate higher levels of fuzziness, uncertainty, and lack of precision and accuracy. Unlike conventional GIS, local communities are intimately involved in the production of neogeographic GIS information, and indigenous forms of knowledge are accepted as valid and important sources of information. With more emphasis on cooperation and collaboration, contemporary GIS practices are considerably more sensitive to issues of access, control, and ownership of spatial information (King 2002, Dunn 2007). The feminist critique of GIS (Schuurman and Pratt 2002),

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传统 GIS

在本质上具有空间决定论性质，因为它要求景观和过程必须以点、线、多边形和像元等几何基本元素来表现。支撑 GIS 的数学拓扑带来了栅格、矢量和对象等数据表征形式。部署这些几何形式，使传统 GIS能够分类自然资源、基础设施、人口模式和环境事件等容易观察的经验现象，却难以处理批判地理学和社会理论所使用的、定义不那么明确的阶级、族群、性别及其他社会身份标记。由于技术塑造我们观看世界的方式，传统 GIS

不可避免地建立并放大了一种基本实证主义哲学视角，以牺牲其他视角为代价。在认识论上，传统制图学和 GIS倾向于把定量和可观察之物置于定性和不可观察之物之上；现实被假定只由可见或可测量的东西构成，事物本身而非事物之间的关系拥有本体论优先性。这种观点坚持演绎的、法则式的理性，把社会建构主义或诠释学等其他逻辑边缘化。传统 GIS因而重视精确、简化、确定性和决定论因果关系，而不是复杂性、模糊性、多重性和偶然性——恰恰是批判社会理论家最关切

的方面。在认识论上，传统 GIS完全依赖真理符合论，特别是二十世纪初维也纳学派形式化的版本：理论陈述的真值由其与观察事实的同构性来判断，即假设与数据是否对应。在这种观点中，真理不加问题地存在于“外部”，不论是否有人相信它；真理不是社会建构，而是客观、既定的事实。真理即使无人相信也依然“真实”。这种观点坚持理论语言与经验语言的严格二分（假定数据不受理论影响：事实是给定的而非制造的），使观察者的作用隐形或无关紧要，并长期支配英语分析哲学和许多社会科学。它把认识和定义社会、自然世界的权力赋予一个不可见、理性、隐含男性化、去情境化、全知的观察者——笛卡尔式“我思”。然而面对强烈批评，许多社会科学家和批判性 GIS
实践者已放弃严格的实证主义世界观，尽管这种观点仍主导物理科学。除了知识观和认识者观，传统 GIS
还依赖一种特定空间观。像区位理论和区域科学的各向同性平原一样，传统 GIS实践者几乎总是诉诸绝对的笛卡尔空间观，把空间视为给定的而非生产的、没有社会起源的同质表面。绝对空间——无限、绝对且同质——在西方历史中源远流长，可追溯至包括几何学在内的古典希腊。以一个统一、连续空间为假设基础的欧几里得几何，在此后两千年主导了空间表征数学。在文艺复兴欧洲，笛卡尔空间观推动了制图、土地测量、土木工程和建筑学的发展，并在文艺复兴与启蒙时代不断复制，成为牛顿物理学的基础。这一视角服务于全球化商品生产与消费的新兴资本主义逻辑。例如，对欧洲航海者来说，把空间平滑地化约为距离使海洋变得可航行，并把地球上纷繁的地方组织成单一、统一、连贯而全景式的世界理解。在美国，乡镇与区段制度发挥了类似作用；在这一过程中，空间被剥夺了具体社会语境，成为抽象线性坐标的有序、统一系统（如经纬度网格）。

for example, reveals both the masculinist bias of conventional GIS and points toward an alternative that is more empathetic and caring in nature by incorporating qualitative measures of emotional states such as fear (Kwan 2002b). Neogeography may be seen as yet one more step in this evolution, initiated by a crowd of citizens who are empowered by W eb 2.0 and social networking technologies (Economist 2010).

Although these practices have substantively altered the ways in which GIS is practiced, we concur with Leszczynski (2009a) that there still remains a deep 'disconnect' between critical human geography' s critiques and many GIS practices. Calls have been made for a critical engagement between social theorists and the community of GIS users on the basis of a common material ontological ground, that is, on the basis of the technology' s applications and not on the disparate, academic linguistic terms that divide practitioners of conventional GIS and social theorists (Leszczynski 2009b, c). W e argue that this disconnection is gradually being overcome by neogeography, thus generating more pluralist ontologies in the age of W eb 2.0.

Neogeography and its consequences

During the past several years, neogeography has enjoyed explosive growth (Turner and Forrest 2008), to the point that Turner (2008) claimed that neogeography may even have assassinated GIS. Neogeography both resembles and departs from close kin such as participatory GIS. Neogeography enables extensive, active bilateral contacts between the producers and analysts of geographic data, resulting in a constantly mutating body of work, usually without a clearly defined political agenda. In contrast, participatory GIS tends to confine local and popular input to the generation of data often tied to controversial issues with explicit political implications, such as local zoning disputes, siting of toxic facilities, and so on (Dunn 2007).

At the core of the shift into neogeography is W eb 2.0, the set of software revolutionizing applications on the web. Key to the architecture of this technology are Asynchronous Javascript And XML (AJAX) and Application Programming Interfaces (API), which allow instantaneous user interactions. The functionality offered by W eb 2.0 has precipitated significant changes from traditional approaches. Whereas conventional GIS is expert-centric, neogeography is obviously user-centric. Most neogeographic practitioners, who harness the technology for their own purposes, often have limited technical skills, and typically care little about academic debates regarding GIS. In this sense, neogeography has helped to foster an unprecedented democratization of geographic knowledge, often with roots far removed from academic experts (Couclelis 2003). Thus Goodchild (2007a, p. 27) maintains that whereas 'the early W eb was primarily one-directional, allowing a large number of users to view the contents of a comparatively small number of sites, the new W eb 2.0 is a bi-directional collaboration in which users are able to interact with and provide information to central sites, and to see that information collated and made available to others.' The interactive websites supported by W eb 2.0, in contrast, allow users to upload locations into online content and apply their data in diverse ways, including, for example, simple displays of locations (e.g.,

favored bird-watching sites) or lists of attributes of a place near a user equipped with a GPS. This approach lies at the heart of mapping websites such as Google Maps, Y ahoo!Maps, OpenStreetMap, and Bing Live Maps. Unlike traditional GIS approaches, users can create, share, and use information through 'crowdsourcing' (Howe 2008), which allow large, widely distributed groups of people to work together toward a common goal. For example, Jackson (2010) reported that OpenStreetMap has attracted 200,000 volunteers annually since its inception in 2004. Similarly, Google Maps was used by 71.5 million people in 2007 and Google Earth by 22.7 million (Haklay et al . 2008). Indeed, rather than relying on state or corporate-produced data through a top-down process, neogeography generates volunteered data and content produced by citizens via a bottom-up process, which has relocated the center of knowledge production from a handful of self-appointed experts to large numbers of people with limited formal geographic training. Sui (2008) labels these changes the 'wikification of GIS,' after Wikipedia, the famously popular, user-generated, online encyclopedia. Wikification represents a significant step forward in the democratization of geographic information, shifting control over the production and use of GIS data from a handful of experts to large groups of users. As a consequence, GIS has increasingly escaped the lonely confines of academia or planning professionals and has been embraced by an enthusiastic and rapidly growing public of amateurs and hobbyists. GIS, it seems, is too important to leave to conventional GIS practitioners.

As a result of its popularity, neogeography' s impacts have multiplied, penetrating the natural sciences, social sciences, health care research, and the humanities. Examples include applications in physical geography (http: //www. climatecollaboratorium.org), human geography (floatingsheep.org, www.popvssoda.com; Elwood 2010), geographical education (Moulder 2008, Papadimitriou 2010), tourism (Girardin et al. 2008), public health (Boulos et al. 2008), and emergency management (Liu and Palen 2010). Neogeography was instrumental in crafting responses to the recent Haitian earthquake and British Petroleum' s disastrous oil spill in the Gulf of Mexico (Zook et al . 2010). Neogeography has also found uses in political geography (Shin 2009; www.fairvote2020.org), popular place descriptions (Dykes et al . 2008), and spatial decision making (Rinner et al . 2008), all of which fall under the broader umbrella of citizen science (Cohn 2008).

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20世纪 90年代，关于传统制图的社会建构主义批评进入 GIS，产生了显著影响。批评者认为，传统 GIS 对社会语境、关系和权力的参与不足，几乎只是把经验主义或实证主义偷运回地理实践的“特洛伊木马”。另一些人担忧 GIS的军事和企业起源与应用、可能侵犯隐私，以及集中式地理空间数据库的全景监控含义。传统 GIS实践者起初以贫乏的认识论自我理解加深了与社会理论家的裂痕，通常只用工程、效率、控制和操纵等纯技术官僚语言来描述 GIS，仿佛它没有社会根源和后果。不过，随着时间推移，GIS社群回应这些批评而持续改变。GIS 完全可以“在非实证主义认识论中运用”。自 20世纪 90年代以来，GIS 的演化产生了一个包含多种新“物种”的奇妙园地，例如批判性 GIS、参与式 GIS、志愿地理学、情感 GIS、定性 GIS、女性主义 GIS、酷儿 GIS、数字地方、民族志与本土知识，以及人文主义 GIScience。这些发展反映了过去十年 GIS 社群与社会理论家、人文学者之间富有成效的互动。GIS科学家与社会理论家的交流促成了所谓批判性 GIS 的出现，试图“使社会像元化、使像元社会化”。尽管这些路径差异显著，它们仍有许多共同点：当代 GIS更关注语境和问题；更重视权力、商品化与监控，也更能容忍模糊、不确定以及不精确和不准确；地方社群直接参与新地理 GIS

信息生产，本土知识形式被承认为有效而重要的信息来源；在更加重视合作与协作的同时，当代 GIS
对空间信息的获取、控制和所有权问题也更敏感。例如，女性主义 GIS的批评揭示了传统 GIS的男性中心偏见，并指向一种更具同理心和关怀性质的替代方案，把恐惧等情绪状态的定性度量纳入其中。新地理学可以被视为这一演化的又一步，由受到 Web 2.0和社交网络技术赋能的公民群体推动。尽管 GIS实践已发生实质变化，批判人文地理学的批评与许多 GIS实践之间仍存在深刻“脱节”。有人呼吁社会理论家与 GIS
用户社群以共同的物质本体论基础展开批判性接触，即依据技术应用，而不是依据分裂传统 GIS
实践者和社会理论家的学术语言。我们认为，新地理学正逐渐克服这种脱节，由此在 Web 2.0
时代生成更加多元的本体论。

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Neogeography, broadly speaking, can be regarded as part of the emerging citizen science, with VGI as its primary input source enabled by the geospatial web (or GeoW eb), that is, the location-based, geocoded data rather than the aspatial information currently hegemonic on the Internet (Scharl 2007).

Although the diverse motivations of people who make contributions to neogeography sites still need further clarification (Cook 2008), neogeography bears distinct similarities to parallel lines of thought such as complexity theory, agent-based modeling, and complex adaptive systems, all of which have also found their way into the academic discipline of geography (O'Sullivan 2004, Manson and O'Sullivan 2006, Batty 2007). Despite these manifest differences, there are also important commonalities. In both neogeography and other approaches, the analytical focus is 'bottom-up,' that is, on the relations among large numbers of actors (rather than the individual actors themselves), who, often inadvertently, create order out of chaos (a view that also resembles Giddens' s structuration theory) and, sometimes, chaos out of order. Through the properties of emergence or self-organization, complexity theory implies that local spatial configurations of interactions affect outcomes at broader systemic scales, or what the literature on the social production of scale calls scale jumping (Smith 2005). Neogeographies are, in a sense, also complex adaptive systems in which conscious agents construct and reproduce systems of meaning, both intended and unintended. Neogeographic practices generally do not follow a trajectory of gradual and linear growth, but indeed often burst into existence suddenly when a critical mass of followers adopt them quickly. Such a development pattern is consistent with the discoveries of the emerging science of networks (Barabasi 2010). Neogeographic systems generate qualitatively more value as their size and complexity grow. Analogously, according to Metcalfe' s law (Hendler and Golbeck 2008), the value of a telecommunication network to each user is proportional to the square of its total number of users. For the geospatial community, the growth from one million GIS users to 400 million neogeographers worldwide is a game-changing shift.

Although neogeography has frequently been celebrated in the popular press in rosy, utopian terms, it is well worth noting that it also suffers from problematic concerns of its own (Kingsbury and Jones 2009). Five issues are noted here. First, simplistic worship of 'crowd wisdom' can easily slip into uncritical reliance upon groups of uninformed amateurs, with potentially disastrous results, or 'crowd stupidity' (Keen 2008, Sunstein 2009). User-generated content tends to rely on polarizing samples of data – only those who either love or hate the subject matter beyond a threshold level tend to post their opinions online. These extremes can drown more subtle and accomodationist interpretations and generate conceptual and political discord (Sunstein 2009). For example, neogeography may amplify simplistic and erroneous stereotypes among selected groups of users, and downgrade the knowledge of experts (Keen 2008). Parks (2009) noted how Google Earth representations of the violence in Darfur often unproblematically reproduced unsophisticated W estern notions about Africa. The popularity of Charles MacKay' s (2010) classic Extraordinary

Popular Delusions and the Madness of Crowds (originally published in 1841) further testifies to the public' s interest in the mantra of crowd wisdom. Second, the glorification of open source software can exact a price in terms of individual creativity and issues of intellectual property rights, such as the music industry' s notorious crackdown on file-sharing sites such as Napster. Third, despite its popular origins, neogeography is not free either from corporate attempts to commodify these technologies and the forms of knowledge they unleash (e.g., Google, Microsoft, Amazon.com). Fourth, there remains an important, largely positivist, epistemological objection: How reliable are the results of neogeographic approaches without random sampling of data, the lynchpin of scientific respectability (Bruns 2007, 2008, Sunstein 2009)? Self-selected groups of users inevitably will create pools of information with profound biases that reflect their social and spatial positionality. For example, the biases of VGI are particularly acute at the global level due to the digital divide (e.g., half of the world' s population has not made even a single phone call, and only one-quarter has access to the Internet). Fifth, neogeography also serves as a potent reminder of the militarization of everyday life by geolocating precisely where everything and everybody is situated on a 24 /7 basis (Kaplan, n.d.). In a world in which powerful corporate and state interests rely upon accurate geocoded data for often unsavory purposes of surveillance and control, VGI may enhance, not reduce, power differentials in the access to and production of geographic knowledge. Perhaps more than ever, we need a broader dialogue on the legal and ethical implications of pervasive penetration of geospatial technologies into our daily lives (Sui 2010).

Neogeography and alternative conceptualizations of space and place

As human geography shifted decisively into various forms of social theory and political economy, absolute space largely gave way to relative and relational conceptions of spatiality. The coincidence of this change with the rise of neogeography is far from coincidental; rather, relational space and neogeographic ways of collecting and representing data may be seen as simultaneously determinant. Just as conventional GIS carried with it a view of space as given and absolute, so too does neogeography offer the possibility of revisualizing space in relational and relative terms (although there are differences between relative and relational space, they are used here synonymously). In contrast to absolute space, which is fixed, asocial, and timeless,

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新地理学及其后果。过去几年，新地理学爆炸式增长，以至于有人宣称它甚至“ 刺杀”了 GIS。新地理学既类似于参与式 GIS，又与之有所不同。它使地理数据生产者与分析者之间形成广泛、积极的双向联系，产生不断变动、通常没有明确政治议程的作品。相比之下，参与式 GIS往往把地方和民众的投入局限于数据生成，并常与地方分区争议、有毒设施选址等具有明确政治含义的争议问题相联系。转向新地理学的核心是 Web 2.0——一组正在革新网络应用的软件。该技术架构的关键是 AJAX 和应用程序编程接口（API），它们支持即时用户互动。Web 2.0的功能带来了相对于传统方法的重大变化：传统 GIS以专家为中心，而新地理学显然以用户为中心。大多数新地理实践者利用技术服务自身目的，技术能力有限，也通常不关心关于 GIS的学术争论。新地理学因此促进了地理知识前所未有的民主化，根基远离学术专家。Gooddhild 指出，早期网络主要是单向的，让大量用户查看少数网站的内容；新的 Web 2.0 则是双向协作，用户可以与中心网站互动、提供信息，并看到信息被汇总后供他人使用。Web 2.0 支持的互动网站允许用户把地点上传到在线内容，并以多种方式应用数据，例如简单显示地点（喜爱的观鸟点），或列出

GPS用户附近某地的属性。Google Maps、Yahoo! Maps、OpenStreetMap 和 Bing Live Maps等制图网站都以此为核心。与传统GIS不同，用户可以通过“众包”创建、分享和使用信息，让广泛分布的大群体共同追求目标。OpenStreetMap 自 2004年创立以来每年吸引约 20 万名志愿者；2007 年 Google Maps 有 7150 万用户，Google Earth 有 2270 万用户。新地理学不依赖国家或企业自上而下生产的数据，而是通过自下而上的公民过程产生志愿数据和内容，把知识生产中心从少数自封的专家转移到大量受过有限正规地理训练的人手中。Sui 将这些变化称为 GIS的“维基化”。维基化把 GIS数据生产和使用的控制权从少数专家转向庞大用户群，是地理信息民主化的重要一步。因此，GIS日益走出学术界或规划专业人士的孤独边界，被热情而迅速增长的业余爱好者群体拥抱。GIS似乎太重要了，不能交给传统 GIS实践者独占。

新地理学的影响随着其普及而扩大，进入自然科学、社会科学、医疗研究和人文学科。应用实例包括自然地理学、人文地理学、地理教育、旅游业、公共卫生和应急管理。新地理学在应对最近的海地地震和英国石油公司在墨西哥湾造成的灾难性漏油事件时发挥了重要作用。它还被用于政治地理学、地方描述和空间决策，这些都属于更广泛的公民科学范畴。广义而言，新地理学可被视为新兴公民科学的一部分；志愿地理信息是其主要输入来源，由地理空间网络（GeoWeb）提供支持，即以基于位置、经过地理编码的数据取代目前互联网上占支配地位的非空间信息。尽管贡献者的多样动机仍有待进一步澄清，新地理学与复杂性理论、基于主体的建模和复杂适应系统等平行思想有明显相似之处，这些思想也都进入了地理学科。在新地理学和其他路径中，分析焦点都是“自下而上”的：关注大量行动者之间的关系，而非行动者个体本身；他们往往无意中从混沌中创造秩序（这一观点也类似吉登斯的结构化理论），有时也从秩序中创造混沌。复杂性理论通过涌现或自组织的性质说明，地方层面的空间互动构型会影响更广泛系统尺度上的结果，这就是社会尺度生产文献所说的“尺度跳跃”。新地理学在某种意义上也是复杂适应系统：有意识的行动者建构并再生产意义系统，包括有意和无意的结果。新地理实践通常不是渐进、线性增长，而是在追随者达到临界规模并迅速采纳时突然爆发式出现。这种发展模式与网络科学的发现一致。新地理系统随着规模和复杂性增长而产生质量更高的价值。按照梅特卡夫定律，电信网络对每个用户的价值与用户总数的平方成正比。对地理空间社群而言，全球 GIS用户从一百万增长到四亿新地理学家的转变，具有改变游戏规则的意义。尽管大众媒体常以光明、乌托邦式语言赞美新地理学，但它也有自身的问题。这里指出五个问题。第一，对“群体智慧”的简单崇拜很容易滑向对无知业余者群体的不加批判依赖，可能带来灾难性后果，即“群体愚蠢”。用户生成内容往往依赖两极化的数据样本：只有极度喜爱或憎恨主题的人才会在线发表意见。这些极端声音会淹没更细腻、寻求调和的解释，制造概念和政治冲突。例如，新地理学可能放大某些用户群体的简单错误成见，同时贬低专家知识。Google Earth 对达尔富尔暴力的表现，就常不加反思地复制了西方对非洲的粗浅观念。查尔斯▲麦凯 1841 年出版的《非同寻常的大众幻觉与群体疯狂》长期流行，也证明公众对群体智慧口号的兴趣。第二，美化开源软件可能以个人创造力和知识产权为代价，例如音乐产业对 Napster 等文件共享网站的严厉打击。

第三，尽管新地理学源自大众，它也无法摆脱企业试图将这些技术及其释放的知识商品化的努力，例如 Google、Microsoft 和 Amazon.com。第四，仍有一个重要的、基本实证主义的认识论疑问：没有对数据进行随机抽样，新地理方法的结果有多可靠？随机抽样是科学可信性通常定义的关键。用户自我选择的群体必然会形成带有深刻偏见的信息池，反映其社会和空间位置。由于数字鸿沟，VGI 的偏见在全球层面尤其严重：世界上一半人口甚至没有打过一次电话，只有四分之一能够上网。第五，新地理学还强烈提醒我们日常生活的军事化：它可以全天候精确标示一切事物和每个人的位置。在强大的企业和国家利益依靠精确地理编码数据进行监视和控制的世界里，VGI 可能不是减少而是加剧地理知识获取与生产中的权力差异。我们或许比以往更需要就地理空间技术普遍渗入日常生活的法律和伦理含义展开广泛对话。新地理学与空间、地方的替代性概念化。随着人文地理学决定性地转向各种社会理论和政治经济学形式，绝对空间大体让位于相对和关系性的空间观念。这一变化与新地理学的兴起同时发生绝非巧合；关系空间以及新地理学收集和表现数据的方式，可以被视为相互决定的。正如传统 GIS携带把空间视为既定且绝对的观点，新地理学也提供了以关系和相对方式重新可视化空间的可能性（尽管相对空间与关系空间有所区别，本文在此同义使用）。与固定、非社会性和无时间性的绝对空间不同，相对和关系空间反映了距离被测量、感知和穿越的多种方式，即空间如何随时间被社会生产和再生产。相对空间描绘的地理不是实证主义那种僵死的几何，而是流动、可变且不断变化的。 新地理学的应用还包括气候协作平台、浮动羊群和流行文化地图、地理教育、旅游、公共卫生及应急管理；它参与了海地地震和墨西哥湾漏油灾害的响应，也用于选举地理、地方描述和空间决策。它与复杂性理论、主体建模和复杂适应系统都把大量行动者的相互关系作为分析起点；局部互动通过涌现和自组织跳跃到更大尺度。新地理系统并非平滑线性增长，而常在用户达到临界质量时突然出现。

relative and relational space reflects the wide varieties of ways in which distance is measured, perceived, and traversed, that is, space as socially made and remade over time. Rather than the long-dead geometries of positivism, relative space portrays geographies as fluid, mutable, and ever-changing.

The genesis of relative and relational space may arguably be traced to the famed seventeenth-century intellectual Gottfried Leibniz (1646–1716), Newton' s rival and the founder of an alternative perspective on spatiality. The Leibnizian view of space is sharply at odds with the prevailing Euclidean /Cartesian emphasis on absolute space. In contrast to Newton, who held that space was absolute in the sense that it existed independently of how it was measured or what it contained, Leibniz held that time and space were comprehensible only with reference to specific frames of interpretation: distance, for example, could only be understood through the appeal of two or more objects situated in space. Relative space, however, remained subordinate to absolute space for many years, at least until the massive time–space compression of the Industrial Revolution reworked the meaning of spatiality itself, as wave upon wave of innovations reduced the friction of distance and led to a widespread reconfiguration of notions of proximity (Warf 2008). Relative and relational space became increasingly intertwined with the dynamics of modern and postmodern capitalism. For example, Kern' s (1983) famous book The Culture of Time and Space 1880–1918 addressed the changing meanings of spatiality in the midst of late nineteenth- and early twentieth-century technological and intellectual changes. The telegraph, telephone, bicycle, photography, automobile, airplane, impressionism, cubism, cinema, and theory of relativity, all blurred conventional meanings of near and far by bringing into contact peoples and societies that had long been isolated from one another. This transformation rendered traditional views of distance, and thus absolute space, less meaningful, and elevated the concept of space as a product of changing social circumstances.

Social theory proved to be decisive in initiating the shift from relative space to relational space, one with explicit social origins and consequences. Relational space, for example, found its way into Marx' s summation of the process by which capitalism reached out and conquered new spaces, enveloping them within ever-broader spatial divisions of labor. Notably, David Harvey' s (1990) famous spatialization of Marxism linked the construction and reconstruction of relational space to capital' s endless search for ever greater profits and outlets for surplus value. In the contemporary historical moment, globalized, postmodern capitalism has given birth to complex geographies in which proximity and transport costs play relatively marginal or even insignificant roles (Sheppard 2002). Airplanes, satellites, television, and telecommunications, including the Internet, for example, steadily led to forms of social interaction in which physical distance is largely rendered irrelevant, even if location and place are decidedly not. Consequently, we have witnessed a series of convoluted landscapes marked by wormholes and tunnels, giving rise to complicated, origami-like spatialities typified by rapidly changing geographies of centrality and

peripherality. Geographic theorizations of relative and relational space include actor–network theory (Murdoch 2006), Massey' s (1993 2005) power-geometries, and Deleuze and Guattari' s (1987) 'schizophrenic' reconceptualization of rhizomes. All of these works point toward a topological conception of space quite at odds with that found in conventional GIS, that is, space as formed by contingent, rapidly changing, interconnected sets of networks in which relational connections among locales rather than their absolute positionality is the dominant characteristic. Neogeography has injected relational space into the hitherto rather closed world of GIS. Egenhofer and Mark (1995) argued that formal models can be developed according to people' s 'naïve geography,' commonsense knowledge about their surroundings that can serve as a means of incorporating daily life and lived experience into the design of GIS (Dourish and Bell 2007, Writer 2009). As cyberspace has become indispensable to ever larger numbers of people, including the 1.9 billion users of the Internet in June 2010 (www.internetworldstats.com /stats.htm), neogeography has allowed innumerable people, most of whom are outside of academia, to chart diverse paths through the virtual metaverse (Smart et al. 2007), the multiple spaces (Li et al . 2010), and 'digi-places' (Zook and Graham 2007b, Graham 2009) that are often populated by avatars (Papagiannidis et al . 2009). According to the report recently produced by the cross-industry public foresight project (www.metaverseroadmap.org), we are rapidly entering a new age of the metaverse – virtual worlds that serve as digital equivalents to the atom-based physical world. The browser for engaging this metaverse will be based upon a three-dimensional web that brings together the following four technologies:

- Mirror worlds – digital representations of non-virtual reality, such as Google Earth, Microsoft' s Virtual Earth, NASA ' s W orld Winds, ESRI' s ArcGlobe, and so on;
- Virtual worlds – digital representations of imagined worlds, such as Second Life and W orld of Warcraft;
- Lifelogging – the self-generated digital capture of information in the real or digital worlds, such as blogs, T witter, Y ouTube, and social interactions via MySpace or FaceBook;
- Augmented reality – sensory overlays of digital information on the real and virtual worlds using head-up displays (HUDs) or other mobile /wearable devices such as cell phones or sensors through participatory sensing.

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相对空间和关系空间的起源，可以追溯到 17 世纪著名思想家戈特弗里德▲莱布尼茨（1646—1716）。莱布尼茨是牛顿的竞争者，也是空间性的另一种观点的创始人。他的空间观与欧几里得 / 笛卡尔式绝对空间的主流强调截然相反。牛顿认为空间是绝对的，独立于测量方式和其中所含事物而存在；莱布尼茨则认为，只有参照特定的解释框架，时间和空间才可理解。例如，距离只能通过诉诸位于空间中的两个或更多对象来理解。多年来，相对空间仍从属于绝对空间，直到工业革命巨大的时空压缩重塑了空间性的意义：一波又一波创新降低了距离摩擦，广泛重新配置了邻近观念。相对空间和关系空间日益与现代、后现代资本主义的动力交织在一起。社会理论在推动从相对空间转向关系空间方面发挥了决定性作用，后者具有明确的社会起源和社会后果。关系空间进入了马克思关于资本主义向外延伸、征服新空间并把它们包入日益扩大的空间劳动分工的概括。大卫▲哈维把马克思主义空间化，将关系空间的建构和重构与资本无止境追逐更大利润、寻找剩余价值出口联系起来。当前全球化、后现代资本主义产生了复杂地理，在其中邻近性和运输成本的作用相对边缘甚至无关紧要。飞机、卫星、电视和包括互联网在内的电信技术，持续促成物理距离基本无关紧要的社会互动形式，尽管位置和地方仍极

为重要。我们由此见证了由虫洞和隧道标记的曲折景观，以及以中心性和边缘性快速变化为特征的、类似折纸的复杂空间性。相对和关系空间的地理理论包括行动者—网络理论、梅西的权力—几何学，以及德勒兹和瓜塔里的“精神分裂”式根茎重新概念化。所有这些著作都指向一种与传统 GIS截然不同的拓扑空间观：空间由偶然、快速变化且相互连接的网络集合形成，地点之间的关系联系而非绝对位置是其主导特征。新地理学把关系空间注入此前相当封闭的 GIS世界。Egenhofer 和 Mark 认为，可以依据人们的“朴素地理学”（关于周围环境的常识知识）发展形式模型，把日常生活和生活经验纳入 GIS设计。随着网络空间对越来越多的人（2010年 6月已有 19亿用户）变得不可或缺，新地理学让无数大多不在学术界的人能够穿行于虚拟元宇宙、多重空间和“数字地方”之中，这些空间常由化身居住。跨行业公共前瞻项目的报告指出，我们正迅速进入元宇宙新时代——作为基于原子物理世界数字对应物的虚拟世界。浏览器将以三维网络为基础，把四项技术结合起来：镜像世界（Google Earth、微软 Virtual Earth、NASA World Wind、ESRI ArcGlobe
等对非虚拟现实的数字表现）；虚拟世界（Second Life、World of Warcraft
等对想象世界的数字表现）；生活记录（博客、Twitter、YouTube 以及通过 MySpace 或 Facebook 进行的社会互动等，对现实或数字世界信息的自我生成式数字捕捉）；增强现实（通过抬头显示器或手机、传感器等移动 / 可穿戴设备，以参与式感知把数字信息叠加到现实和虚拟世界）。 工业革命的时空压缩降低了距离摩擦，重新定义了邻近性；电报、电话、自行车、摄影、汽车、飞机、印象派、立体主义、电影和相对论，使长期隔绝的人群相互接触，模糊了“近”与“远”。马克思主义和哈维的空间化把关系空间与资本扩张、空间劳动分工及剩余价值联系起来。全球化资本主义形成由虫洞、隧道和中心 / 边缘快速变动构成的复杂景观；行动者—网络理论、梅西的权力几何学及德勒兹—瓜塔里的根茎概念都强调这种拓扑空间。

Viewed from the perspective of this emerging metaverse, traditional GIS has focused almost exclusively on georeferenced Cartesian coordinates defined by Euclidean geometry, a view that reifies absolute conceptualizations of space. In contrast, the diverse approaches grouped under the umbrella of neogeography include elements from all these four worlds that are linked together through heterogenous geo-tags, an approach in accordance with a relational view of space (Jensen 2010). Jones (2009) holds that relational thinking is a paradigmatic departure from the concerns of absolute space because it dissolves the boundaries between objects and space, that is, it sutures phenomena and context together into inseparably intertwined entities. Such a stance is firmly in keeping with the relational view of space, in which space does not exist as an entity in and of itself, over and above material objects and their spatial–temporal relations and tensions.

In addition to promoting an alternative view of space, neogeography also entails the incorporation of place with an emphasis on local information (Willis 2010), another conceptual shift that is quite different from traditional GIS practices. According to Tuan (1977), whereas space has often been associated by geographers with freedom, movement, distance, potential, and abstraction, place often implies confinement, stability, proximity, meaning, and the concrete. As Sack (1980) and Casey (1998) cogently argued, conceptualizations of space and place have a rich history and do not lend themselves to easy summary. Such a maneuver allows neogeographers to incorporate hitherto neglected data such as personal feelings at particular places and times (http: //www.wefeelfine.org). Many neogeographic practices are in fact detailed descriptions of particular places or regions. Graham (2009) resurrected the once-popular palimpsest metaphor to describe what neogeographers have been contributing online: W eb 2.0 technologies essentially serve as the new medium, similar to the papyrus- or parchment-based documents of an earlier era, for us to write and rewrite layers of information about the world. W e need the concepts of both space and place to comprehend the ontology of neogeography.

Thus, rather than a single, uniform plain over which social relations unfold, the ontology of neogeography tends to be partial, overlapping, and specific to the interests of particular groups of users. There exists not one, single, coherent neogeographic space, but a vast multiplicity of user-generated places specific to particular communities of interest. Such places, generated through countless bottom-up interactions of users who are widely dispersed among many physical locations, are virtual, constantly changing, and often have only tenuous linkages to material landscapes. In addition, by allowing the local to become global, neogeographies cross conventional geographic scales with ease. In this sense, neogeography is compatible with the Deleuze–Guattarian ‘flat ontology,’ or spatialities unmarred by the obfuscating effects of scale (Marston et al . 2005). This notion is demonstrated by the multiple, multifaceted examples in the book Else/Where: Mapping New Cartographies of Networks and Territories (Abrams and Hall 2006), which embody the relational view of space and palimpsest view of place.

Neogeography and theories of truth: epistemological

anarchism in the making?
If neogeography has offered alternative, pluralist ontologies of space and place, it also has important epistemological consequences. In particular, as diverse groups of people with varying agendas harness the web to upload their own data and interpretations, conventional views of what constitutes valid knowledge and truth come into question. For example, groups utilizing neogeographic technologies are not likely to generate random samples of data, a criterion that lies at the heart of commonly accepted (i.e., positivist) definitions of 'the scientific method.' Rather, the digital inputs that define group-specific, user-generated virtual worlds are selected with the commonality of intent that unites them into communities of interest. User-generated maps of outstanding nudist sites, endangered bird observations, handicapped-accessible restrooms, optimal bike paths or camping locations, accident-prone roads, green buildings, good shoe stores, vegetarian restaurants, or where people were located on 11 September 2001 may not pass the traditional standards of journal peer review and may not correspond with the 'facts' at the 95% statistical confidence level, but their efforts yield results that are important and meaningful to their respective groups of contributors, creating a 'people' s geography' quite distanced from the rarified world of academia. What, then, is 'true' when people, particularly nonexperts, generate their own data and stories to make sense of the world in ways that may be markedly at odds with the rigid criteria demanded by GIS experts?
Departing from the correspondence theory of truth, neogeography allows for epistemological communities to construct 'truths' relative to specific groups of users. What is 'true,' for example, of place-based stories uploaded by a group of students through their iPhones may be quite different from the truth of environmental activists seeking to define the habitat of an endangered minnow. Such a perspective clearly points to the social and spatial construction of knowledge and the criteria by which we judge it to be 'true'. Truth, in this context, is removed from the exclusive domain of the intellectual, freed from its moorings in the correspondence theory, and instead is generated by those to whom it is useful in pragmatic, everyday terms.
There are many alternatives to the correspondence theory of truth, including constructivist, coherence, pluralist, performative, and pragmatist perspectives, all of which, to varying degrees, depart from conventional (i.e., orthodox positivist or 'scientific') perspectives by pointing to the

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从这个正在形成的元宇宙视角看，传统 GIS几乎只关注由欧几里得几何定义的地理参照笛卡尔坐标，从而把绝对空间概念实体化。相反，以新地理学为总称的多样方法包含四种世界的要素，并通过异质的地理标签把它们连接起来，这与关系空间观一致。关系性思维之所以是对绝对空间关注的范式性背离，是因为它消解了对象与空间之间的边界，把现象和语境缝合为不可分离、相互交织的实体。关系空间观认为，空间并不是凌驾于物质对象及其时空关系和张力之上的、独立存在的实体。除了推动替代性的空间观，新地理学还强调纳入地方和本地信息，这同样不同于传统 GIS。Tuan 指出，地理学家常把空间与自由、移动、距离、潜能和抽象联系起来，而地方往往意味着限制、稳定、邻近、意义和具体。空间与地方的概念化有着丰富历史，不能轻易概括。新地理学由此可以纳入此前被忽视的数据，例如特定地点和时间的个人感受。许多新地理实践其实是对特定地点或区域的详细描述。Graham
重新启用曾经流行的“重写本”隐喻来说明新地理学家在线贡献的内容：Web 2.0技术成为新的媒介，如同早期以纸莎草纸或羊皮纸为基础的文献，让我们不断书写和重写有关世界的信息层。要理解新地理学的本体论，我们需要空间和地方这两个概念。因此，新地理学的本体论不是社会关系展开其上的单一、均质平面，而是部分的、重叠的，并具体对应特定用户

群体的利益。不存在一个单一、一致的新地理空间，而是存在与特定兴趣社群相对应、由用户生成的巨大多样地方。这些地方由分散在许多物理位置的用户自下而上的无数互动生成，是虚拟且不断变化的，与物质景观的联系往往十分脆弱。此外，通过让地方性变成全球性，新地理学轻易跨越传统地理尺度。在这一意义上，新地理学与德勒兹—瓜塔里式“扁平本体论”相容，即空间性不受尺度遮蔽效应的损害。《Else/Where：网络与疆域的新制图》一书中的多面案例说明了这一点，它们体现了关系空间观和地方的重写本观。新地理学与真理理论：认识论无政府主义正在形成？如果新地理学提供了空间与地方的替代性多元本体论，它也产生了重要的认识论后果。当具有不同议程的群体利用网络上传自己的数据和解释时，关于何为有效知识和真理的传统观念便受到质疑。例如，使用新地理技术的群体不太可能生成随机数据样本，而随机抽样是通常接受的（实证主义）“科学方法”的核心标准。定义特定群体用户生成虚拟世界的数字输入，是按照共同意图选择的，这种共同意图把他们联合为兴趣社群。 元宇宙四项技术的具体含义是：镜像世界把 Google Earth、Virtual Earth、NASA World Wind 和 ArcGlobe 等现实数字化；虚拟世界把 Second Life、World of Warcraft
等想象世界数字化；生活记录把博客、Twitter、YouTube 及社交互动中的信息自我捕捉下来；增强现实则利用抬头显示器、手机和参与式传感器把数字信息叠加在现实和虚拟世界上。新地理学的地方性实践还包括记录人在特定地点和时刻的情绪，以及由用户不断覆盖、改写的“重写本”式地方。

social origins and applications of knowledge. Whereas far from constituting some homogenous whole, these perspectives share a notion of ‘truth’ as something rather more complex than an unbiased representation of objective reality independent of the observer and applicable in all times and places. As Gibson-Graham (2000, p. 105) notes, ‘If knowledge is not assigned the task of providing an accurate reflection of reality . . . then research doesn’t simply reveal “what’ s out there” in the world.’ Instead truth is considered as a partial, authored, and embodied series of intertwined statements created by diverse groups of people with widely different incentives, for numerous contrasting purposes, in specific and widely varying historical and geographic contexts (Livingstone 2003, Finnegan 2008). In this reading, knowledge is always embodied and spatially situated. As Hardy (2010) demonstrated, for example, even the number of contributors for Wikipedia entries follows a distance-decay function – people tend to contribute more information about places closer to where they live. However, in neogeography, the sharp divisions between the knower and the known, representations and the world they portray, epistemology and ontology are deliberately and creatively blurred, and truths (for there is always more than one) are repositioned as a partial, contingent series of statements that reflect lived reality and are useful in it. This line of thought owes much to John Dewey’ s and particularly William James’ s (1907, 1909) pragmatist epistemology, in which ‘truth’ is determined and confirmed by its utility and effectiveness in application, that is, from its consequences; thus, ‘the “true” is only the expedient in our way of thinking, just as the “right” is only the expedient in our way of behaving’ (James 1909, p. 2). Rather than the panoptic gaze of professional GIS practitioners, neogeography has facilitated a democratization of truth production, transferring its creation and application from experts to the masses. To explicate this point, the works of three important philosophers are noted here: Paul Feyerabend, Jürgen Habermas, and Richard Rorty.

Feyerabend (1975) was instrumental in dethroning the correspondence theory of truth, advocating what amounted essentially to an anarchist theory of knowledge and ontological pluralism. In this line of thought, which preceded and facilitated the emerging discourses of postmodernism, there is no single observable reality or one methodological style. Rather, there are multiple realities, many different and equally viable ways of constructing knowledge, all of which require an enormous tolerance for inconsistency, incompleteness, and uncertainty. The parallels between Feyerabend’ s views and the perspectives of applied neogeographers are difficult to deny: both hinge on the acceptance of diverse, incommensurable ways of knowing, an acknowledgement of the multiple ways in which knowledge and social practices are intertwined, and a democratic celebration of very different bodies of knowledge outside the purview of self-appointed experts.

Another important contribution, and one of considerable relevance to neogeography, is Habermas’ s consensus theory of truth. As Habermas (1989) argues, communications are central to the social process of consensus and truth construction, through which individuals and communities of interest partake in the public, discursive interpretation

of reality (cf. Calhoun 1992). Habermas' s 'ideal speech situation' consisting of unfettered discourse is central to the 'public sphere' in which social life is constructed and reproduced and through which truth is constructed in the absence of barriers to communication. Truth, interpreted in this way, is inseparable from lived experience, intent, and social practice. In this reading, all participants in a debate theoretically have equal rights and abilities to make their views known and to challenge any other view. When all power relations have been removed from the freedom to engage in discourse, the only criteria for resolving contesting claims is their truth-value; importantly, 'the participants in an ideal speech situation [must] be motivated solely by the desire to reach a consensus about the truth of statements and the validity of norms' (Bernstein 1995, p. 50).
If one adopts a Habermasian approach in which democracy is approximated by an ideal speech situation' of unfettered discourse in the public sphere, neogeography provides a reasonably good approximation. Thus, neogeography practitioners can actively provide input into geographic representations, challenge dominant views, and mold the outcomes toward their own ends (Coleman et al . 2009, Graham 2009, Liu and Palen 2010). Accordingly, neogeography makes it possible for grassroots and nongovernmental organizations to shape the technology to their own needs (Sieber 2000). The explosion of cellular phones and similar portable devices has greatly facilitated the ability of many groups to upload and download neogeographic data (Ling and Campbell 2009). As Schuurman (2009, p. 571) observed, 'W e face scenarios, previously unimaginable, of human beings as sensors gathering constant information with their cell phones and reporting back information to central, web based data collating sites.'
Dunn (2007) cautions that participatory GIS, like neogeography, nonetheless faces real constraints to users, including the technical skills needed to upload data. Moreover, simply providing input is not the same as shaping policy outcomes that result from utilizing geodata. Of course, access to neogeography technologies is limited by social factors such as income, education, age, ethnicity, and gender, all of which are significant determinants of the 'digital divide' as well (Crang et al . 2006). Nonetheless, inasmuch as anyone with simple access to web tools to upload data and download results can participate in neogeographic communities, Habermas' s ideal speech situation is largely realized for a vast potential pool of participants.
Similarly, Rorty (1979) offered a famous shift in metaphors for understanding knowledge. If the mirror served as the ideal 'reflection' of knowledge construction

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真理理论的替代方案包括建构主义、融贯论、多元论、施为论和实用主义观点；它们都在不同程度上偏离正统实证主义或“ 科学”观点，指出知识的社会起源及其应用。这些观点并非同质整体，却共享一种看法：真理比“ 不受偏见、独立于观察者、在一切时间和地点都适用的客观现实再现”复杂得多。正如 Gibson-Graham 所说，如果知识不再承担准确反映现实的任务，研究就不只是揭示世界上“ 有什么”。真理应被视为由具有不同动机的人群，在具体且多变的历史和地理语境中，为相互差异的目的创造的、部分的、有作者和身体经验的相互交织陈述。知识总是具身的，并在空间上处于特定位置。Hardy 的研究显示，甚至维基百科条目的贡献者数量也遵循距离衰减函数：人们往往更多贡献与居住地较近的地方有关的信息。新地理学有意而富有创造性地模糊认识者与被认识者、表征与所描绘世界、认识论与本体论之间的界线；真理（因为真理总不止一个）被重新定位为反映生活现实且在其中有用的、部分的和偶然的陈述系列。这种思路深受约翰▲杜威、尤其是威廉▲詹姆斯的实用主义认识论影响：真理由其在应用中的效用和有效性、也就是由后果来确定和证实；“ 真”只

是我们思考方式中的权宜之计，就像“ 正确”只是我们行为方式中的权宜之计。与专业 GIS实践者的全景式凝视不同，新地理学推动了真理生产的民主化，把真理的创造和应用从专家手中转移给大众。费耶阿本德帮助推翻真理符合论，主张近乎无政府主义的知识理论和本体论多元主义。在这一先于并促成后现代主义的话语中，不存在单一可观察现实或唯一方法风格，而存在多重现实以及许多同样可行的知识建构方式；所有方式都要求对不一致、不完整和不确定性具有巨大容忍度。实际新地理学家的观点与费耶阿本德高度相似：二者都接受多样且不可通约的认识方式，承认知识与社会实践有多种交织形式，并民主地赞美自封专家管辖范围之外的不同知识体系。哈贝马斯的真理共识论同样与新地理学密切相关。他认为，交流是共识和真理建构的社会过程核心；个人和兴趣社群通过公共话语参与对现实的解释。“ 理想言说情境”中的不受阻碍话语，是建构和再生产社会生活的公共领域核心，真理正是在没有沟通障碍时建构的。所有辩论参与者在理论上都有平等权利和能力表达观点、质疑其他观点；当权力关系不再妨碍话语自由时，解决相互竞争主张的唯一标准就是其真值。理想言说情境中的参与者必须只受达成陈述真理与规范有效性共识的愿望驱动。若把民主理解为公共领域不受阻碍话语的理想言说情境，新地理学提供了相当好的近似。实践者可以对地理表征主动提供投入、挑战主流观点，并按照自身目标塑造结果；基层组织和非政府组织能够使技术服务自身需要。手机及类似便携设备的爆炸式普及，使许多群体上传和下载新地理数据更加容易。我们面对着过去无法想象的情景：人类成为传感器，用手机持续收集信息，并向中央的网络数据汇总站回报。参与式 GIS和新地理学仍受到技术技能、政策影响力及数字鸿沟的限制；收入、教育、年龄、族群和性别等社会因素决定了技术可及性。不过，只要能使用网络工具上传数据并下载结果，任何人都可参与新地理社群，哈贝马斯的理想言说情境便在很大程度上为潜在的广大参与者实现。

from the Enlightenment onward (with its emphasis on vision, accuracy, and light), then the conversation serves as the perfect vehicle to summarize the forms of post-structuralist knowledge construction in the present era – a messy series of dialogues in which each voice is partial, incomplete, embodied, and contingent. In geography, a challenge to the hegemony of the visual has gradually occurred as the field has come to embrace social constructivist versions of truth rather than correspondence theories (Sui 2000). Neogeography is thus much more of a 'conversation' than it is a 'mirror' of the world in which the truths constructed are relative and useful for specific communities, there is no value-free arbiter to decide what is accurate/true, and accuracy is decided by consensus and pragmatic value. Moving away from the correspondence theory of truth, the conversational view puts more emphasis on practice, performance, and the speech acts that lead to points of agreement between contending worldviews (Strawson 1997). Neogeography has edged us closer to the performative theory of truth with an emphasis on actual spatial practices, that is, 'performing' these geographical or cartographic tasks on a sustained, repeated basis. Instead of truth, 'performativity' or 'truthiness' has been pushed to the front, that is, the quality of an idea 'being done right' or 'feeling true' without voluminous empirical evidence (Manjoo 2008). This line of thought has distinct, albeit as yet largely unselfconscious, parallels to nonrepresentational theory, with its emphasis on affect and prelinguistic, precognitive forms of knowledge (Thrift 2008).

Through selective exposure and peripheral processing, neogeographic media have enabled large numbers of people to pick and choose those sources of information that mesh conveniently with their ideological presuppositions. There are obviously both advantages and disadvantages to this approach. Most notably, it is not always the case that the 'wisdom of the crowd' is superior to that of a few experienced individuals. By utilizing data that only confirm their beliefs, users may never be confronted with disturbing or contradictory sources of information, which may reinforce existing stereotypes. In this sense, neogeography runs the risk of mirroring the highly specialized media markets in which viewers tend to utilize sources of news that conform to their political prejudices (Schuman 2004). However, by putting massive amounts of data and powerful visualization tools firmly in the hands of users and consumers rather than an elite group of expert producers, neogeography forces a broader recognition of the degree to which truth values reflect broader social concerns such as trust, reputation, and the credibility of volunteered information (Bishr and Kuhn 2007, Bishr and Mantelas 2008, Flanagin and Metzger 2008). Accuracy, in this context, is largely a matter of ideology and preference, contingent upon context and purpose, and tailored to the specific interests of communities of interest. In facilitating the emergence of numerous 'neighborhoods of truth,' neogeography encourages us to abandon the ridiculous holy grail of universal generalizations and come to terms with the place-bound nature of geographic knowledge. In its stead, it opens up venues for viewing the world through the eyes of particular groups bound together by lifestyle, political values, recreational habits, and other dimensions

of social life.

Conclusions

The year 2010 marks the 20th anniversary for the coining of the term ‘geographic information science’ (Goodchild 2010). During the past 20 years, the GIScience community has made enormous progress in addressing the fundamental issues related to geographic data representation, analysis, and visualization (Goodchild 2009b). The explosive growth of neogeography in recent years poses new challenges and also offers new opportunities for both GIScientists and geographers. In this article, we aim to set the stage for a wider dialogue concerning the broader implications of neogeography, especially the ontological implications of neogeographical practices regarding the alternative conceptualizations of space and place and their consequences for theories of truth.

The recently released National Research Council (2010) report on the geographical sciences in the United States identified citizen mapping and mapping citizens as 1 of the 11 strategic directions on which geographers should focus in the near future. Citizen mapping and mapping citizens fit squarely within the scope of neogeography. The convergence of expert-led basic GIScience and geographic research with citizen-initiated neogeography is one area that likely will lead to exciting breakthroughs in GIScience and geography. As Downs (2010) recently observed, geography has been a popular subject throughout the history of the discipline, yet professional geographers have not done an adequate level of job of communicating their research to a broader audience. W eb 2.0 technologies have brought the crowd of spatial data users to us through neogeography, and geographic knowledge is increasingly no longer the province of geographers and GIScientists, if it ever was. It is imperative for professional geographers and GIScientists to seize this unprecedented opportunity to make geography and GIScience more socially relevant by acknowledging the validity of user-generated communities of truth, the multiplicity of criteria that define useful knowledge, and the complex spatialities that arise from crowds who harness GIS for their own, rather than academic, purposes.

The popularity of neogeography reflects the increasingly fractalized nature of social and spatial life under contemporary capitalism, a world in which the structure and meaning of geography have been profoundly transformed under the simultaneous impacts of globalization and increasingly democratized information technologies.

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新地理媒体通过选择性接触和外围处理，使大量人群能够挑选那些与其意识形态预设相契合的信息来源。这种方法显然有利有弊。尤其要注意，“ 群体智慧”并不总是优于少数有经验的个人。只使用确认自身信念的数据，用户可能永远不会遇到令人不安或相互矛盾的信息，从而强化既有成见。新地理学因此有模仿高度专业化媒体市场的风险，观众倾向于选择符合政治偏见的新闻来源。然而，当新地理学把海量数据和强大的可视化工具交给用户和消费者，而不是精英专家生产者时，它迫使人们更广泛地承认：真值反映信任、声誉和志愿信息可信度等社会关切。在此语境中，准确性主要是意识形态和偏好的问题，取决于情境与目的，并按照兴趣社群的具体利益加以调整。新地理学促成众多“ 真理邻域”的出现，鼓励我们放弃普遍概括这一荒谬的圣杯，承认地理知识受地方束缚的性质；它让我们能够通过由生活方式、政治价值、休闲习惯及其他社会生活维度联系在一起的特定群体之眼来观看世界。结论。2010

年是“ 地理信息科学”这一术语提出二十周年。过去二十年，GIScience

社群在处理地理数据表征、分析和可视化等根本问题方面取得了巨大进展。近年新地理学的爆炸式增长，为 GIS科学家和地理学家带来新的挑战与机遇。本文旨在为更广泛的对话奠定基础，讨论新地理学的广泛含义，尤其是新地理实践对空间

和地方替代性概念化的本体论影响，以及它们对真理理论的后果。美国国家研究委员会最近发布的地理科学报告，把公民制图和为公民制图列为地理学家近期应关注的 11 个战略方向之一，这正属于新地理学的范围。专家主导的基础 GIScience 和地理研究与公民发起的新地理学相汇合，很可能在 GIScience 和地理学中带来令人兴奋的突破。地理学一直是大众喜爱的学科，但专业地理学家没有充分把研究传达给更广泛的受众。Web 2.0
技术通过新地理学把空间数据使用者的群体带到我们面前；地理知识越来越不再是地理学家和 GIS
科学家的专属领域（如果它曾经是的话）。专业地理学家和 GIS科学家必须抓住这一前所未有的机会，通过承认用户生成的真理社群的有效性、定义有用知识的多重标准，以及群众为自身而非学术目的运用 GIS
所产生的复杂空间性，使地理学和 GIScience 更具社会相关性。新地理学的流行反映了当代资本主义下社会和空间生活日益碎片化的性质：在全球化与信息技术日益民主化的同时冲击下，地理的结构和意义被深刻改变。由于允许用户建构自己的地理表征，新地理学可能十分适合这一历史时刻强烈的时空压缩，或适合斯文格多夫所理论化的、嵌入易变后现代“超空间”的区域。简言之，位置越来越成为生产与协商的结果，而非先验给定。如果后现代社会呈现出一种“纤维状、线状、金属丝状、细绳状、绳索状、毛细管状”的特征，永远无法由层级、层次、领土、领域、类别、结构或系统等概念捕获，那么新地理学似乎就是普通公民向自己、为自己、并常常由自己表现其地理——他们建构并生活其中的空间性——的理想方式。致谢：感谢 Jay Knox 和 Wenqin Chen 提供研究协助。参考文献：原文所列作者、年份、期刊、卷页和网址均完整保留；书名和文章题名的中文含义如下：Abrams 与 Hall《无处 / 何处：网络与疆域的新制图》；Aitken 与
Craine《情感视觉地理与 GIScience》；Armstrong 与
Ruggles《地理信息技术与个人隐私》；Barabá si《爆发：我们所做一切背后的隐藏模式》；Barnes 与 Sheppard《“没有什么包含一切”：走向英语经济地理学的参与式多元主义》；Batty《城市与复杂性》；Bernstein《恢复伦理生活：哈贝马斯与批判理论的未来》；Bishr 与 Kuhn《自下而上的地理空间信息：信任与语义问题》；Bodenhamer 等《空间人文：GIS 与人文学术的未来》；Boulos 等《实践中的 Web GIS VI：面向公共卫生新地理学家的地理混搭演示》；Brown 与 Knopp《使地图酷儿化：碰撞认识论的生产性张力》；Bruns《未来由用户主导：走向广泛的生产使用》及《博客、维基百科、第二人生及其他：从生产到生产使用》；Calhoun《哈贝马斯与公共领域》；Casey《地方的命运：一部哲学史》；Cohn《公民科学：志愿者能做真正的研究吗？》；Coleman
等《志愿地理信息：生产者的性质与动机》；Cook《贡献革命：让志愿者建设你的企业》；Coote 与
Rackham《新地理数据质量——这是问题吗？》；Cope 与 Elwood《定性
GIS：混合方法路径》；Cosgrove《地理与视觉：观看、想象和表现世界》；Couclelis《不确定性的确定性：GIS 与地理知识的界限》；Crampton 与 Krygier《批判制图学导论》；Crang
等《连接的可变几何：城市数字鸿沟与信息技术的使用》；Deleuze 与 Guattari《千高原：资本主义与精神分裂》；Dobson 与 Fisher《全景监狱的变动地理》；Dodge 与 Kitchin《代码与空间的转导》；Dourish 与
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Mark《朴素地理学》；Elwood《参与式 GIS的批判问题：解构、重构与新的研究方向》。罗蒂提出了理解知识的隐喻转换：启蒙时代以来，镜子象征准确、明亮的知识反映；后结构主义时代则应以对话来理解知识——一连串混杂的谈话，每种声音都部分、不完整、具身且偶然。新地理学更像世界的“对话”而不是“镜子”：真理对特定社群而言是相对且有用的，没有价值中立的裁判，准确性由共识和实用价值决定。施为性强调持续、重复地“做”地理或制图任务；“真感”（truthiness）指一个观念无需大量经验依据，却让人感觉做得对或像是真的。这与非表征理论对情感、前语言和前认知知识的强调相呼应。

Because it allows users to construct their own geographic representations, neogeography may be well suited to the intense time–space compression of this historical moment, or what Swyngedouw (1989) theorized as regions embedded in a volatile, postmodern 'hyperspace.' Location, in short, has become increasingly a matter of production and negotiation rather than being given ap r i o r i. If postmodern society exhibits a 'fibrous, thread-like, wiry, stringy, ropy, capillary character that is never captured by the notions of levels, layers, territories, spheres, categories, structures, or systems' (Paasi 2004, p. 541), then neogeography seems to be an ideal way for ordinary citizens to represent their geographies – the spatialities they construct and in which they live – to themselves, for themselves, and often by themselves.

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结论的后半部分指出，新地理学允许用户构建自己的地理表征，因此适合当下剧烈的时空压缩和后现代“超空间”。位置越来越是生产和协商的结果，而非先验给定；后现代社会呈现纤维、线、丝、绳和毛细管般的特征，无法由层级、领土、类别、结构或系统等概念完整捕捉。新地理学因而成为普通公民向自己、为自己、常常由自己表现其地理和空间生活的理想方式。致谢：感谢 Jay Knox 和 Wenqin Chen 的研究协助。英文原文随后列出完整参考文献（作者、年份、题名、期刊、卷期、页码、出版社、DOI、网址及访问日期）；这些条目在本双语文件中全部保留，题名的中文翻译见第 11—13 页译文。

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